American Computer Science League

2023-2024 ● Contest 2: Short Problems Solutions ● Senior Division

1. Prefix/Infix/Postfix Notation

* / + ^ 3 2 6 5 * 4 ^ 2 / / 8 2 2 = * / + (^ 3 2) 6 5 * 4 ^ 2 / (/ 8 2) 2
= * / (+ 9 6) 5 * 4 ^ 2 (/ 4 2)
= * (/ 15 5) * 4 (^ 2 2)
= * 3 (* 4 4)
= * 3 16
= 48

E. 48

2. Prefix/Infix/Postfix Notation

5 ! 2 3 − | * 2 ! / 2 3 ^ | + = (5 !) (2 3 −) | * (2 !) / (2 3 ^) | +
= 120 (-1 |) * 2 / (8 |) +
= (120 1 *) 2 / 8 +
= (120 2 /) 8 +
= 60 8 +
= 68

D. 68

3. Bit-String Flicking

(LSHIFT-2 01101) AND (RCIRC-1 (LSHIFT-2 (NOT 11010)))
= 10100 AND (RCIRC-1 (LSHIFT-2 00101))
= 10100 AND (RCIRC-1 10100)
= 10100 AND 01010
= 00000

D. 00000

4. Bit-String Flicking

Let X = abcde

LHS = (RSHIFT-2 01110) OR X AND (LCIRC-1 (LSHIFT-1 10110))
= 00011 OR abcde AND (LCIRC-1 01100)
= 00011 OR (abcde AND 11000)
= 00011 OR ab000
= ab011
RHS = 01011

LHS = RHS ⇒ab011 = 01011 ⇒ a = 0, b = 1, c = *, d = *, e = *
Therefore X = 01***

B. 01***

5. LISP (SETQ X '(1 2 (a (b) c) ((1 a) (2 b) c))) (CDADDR '(1 2 (a (b) c) ((1 a) (2 b) c))) = (CDADR '(2 (a (b) c) ((1 a) (2 b) c))) = (CDAR '((a (b) c) ((1 a) (2 b) c))) = (CDR '(a (b) c)) = ((b) c)	C. ((b) c)